

Phase 1 Day-1 Diagnostic Sprint: H2 Diagnostic Positive on Amazon Clothing

An In-Depth Analysis of the `return code 1` Outcome
and Its Implications for the rev44 Phase 1 Stop-Gate Verdict

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Bottom line. The non-zero return code reported at the end of the ~12-hour Day-1 sprint is *not* an infrastructure failure. It is the **intended diagnostic-positive exit code** for hypothesis **H2** defined in the Phase 1 Root-Cause Roadmap, §2.1. Because `meta_categories.npy` is absent on disk, `utility/load_data.py` silently fell back to a deterministic hash and **every** APC-driven training run, including all 20 runs of the Phase 1 four-variant sweep in cell 16, was executed on phantom clusters. The Phase 1 stop-gate verdict is therefore **confounded by H2** and must be re-evaluated after the real metadata file is restored.

1 TL;DR

- **Return code 1 $\neq$ an operational error.** It is precisely the **H2 SUSPECTED** signal from roadmap §2.1 – the Day-1 script is telling you: “*the metadata file is missing; APC has been running on a hash-based fallback from epoch 0.*”
- This is a **positive diagnostic** for H2 among the ten root-cause hypotheses in the roadmap, and it **automatically invalidates a portion of the Phase 1 conclusions**, because all 20 multi-seed training runs in cell 16 also executed with the same missing-meta configuration – meaning APC produced phantom signals in **every** variant.

2 Interpreting the Warning Banner

```
[pre-flight] MISSING ...\data\Clothing\meta_categories.npy
return code: 1
[WARN] Non-zero exit code -- review logs above.
```

Cross-referenced against the source of `day1_diagnostic_sprint.py` (lines 106–113):

```
if not exists:
    report["h2_suspected"] = True
    report["warnings"].append(
        "meta_categories.npy MISSING -> load_data.py uses deterministic hash "
        "fallback (phantom APC clusters; roadmap H2)."
    )
```

and lines 305–306:

```
if rep["h2_suspected"]:
    exit_code = 1
```

⇒ The exit code 1 here is **not a crash**; it is the “diagnostic-positive” exit code that the roadmap deliberately uses for **fail-fast** signalling. D2, D3 and the APC-off probe **still run after D1** (see lines 310, 340 and 370 of the script – none of those branches is short-circuited by a `return`).

Natural consequences

- The ~ 12 -hour wall-clock run almost certainly completed D2 (4 values of K , 1 run each) + D3 (1 run) + the APC-off probe (1 run) = **6 GPU runs total**.
- Only the opening banner and the closing return code were pasted – the body of stdout, which contains the D2/D3/APC results, was truncated during the copy.
- The full diagnostic conclusion requires reading the per-run log files (see the **Action items** section below).

3 Serious Implication for the Phase 1 Sweep (Cell 16)

The Phase 1 sweep in cell 16 (20 runs: 5 seeds $\times$ 4 variants) was executed **before** cell 15 on the **same machine** with the **same data directory**. Because `meta_categories.npy` did not exist, `utility/load_data.py` silently entered its hash-fallback branch for **all 20 runs**. The per-variant impact is summarised in Table 1.

Variant	damps_apc	Hash-fallback?	Measured R@20	Revised verdict
(a) anchor	=1	Yes (APC on, fake clusters)	0.0795 ± 0.0025	APC signal contaminated
(b) tau03	=1	Yes	0.0829 ± 0.0007	APC signal contaminated
(c) avrf_off	=1	Yes	0.0803 ± 0.0021	APC signal contaminated
(d) combined	=1	Yes	0.0851 ± 0.0012	APC signal contaminated

Table 1: Per-variant H2 contamination assessment for the Phase 1 sweep.

Warning. All four variants run with `-damps_apc 1` by default (see the `Namespace` echoed in the first line of cell 16: `damps_apc=1`). Consequently:

The result $R@20 = 0.0851$ for the **combined** variant **is not the true ceiling of the recipe**. It is the recipe *plus* phantom-APC noise. The real gap versus the paper’s 0.0881 could be smaller *or* larger than the reported -3.4% ; nothing definitive can be concluded until either (i) the true metadata file is restored, or (ii) APC is disabled entirely.

This directly explains the stop-gate verdict produced by cell 20: **all four variants FAIL Phase 1** with $R@20 < 0.0870$, which is exactly the qualitative signature of H2 (hash-fallback weakens the cluster signal).

4 Relationship to the Roadmap’s Top-3 Root Causes

The positive H2 verdict **raises the priority** of two actions:

1. Restore the original `meta_categories.npy` (from the MMHCL paper release) and re-run the Phase 1 sweep. This is a **necessary precondition** for any further Phase 1 conclusion to be valid.
2. Read the D2/D3 outputs that have already been computed (but not yet pasted) to decide whether K needs to be increased *after* the metadata is restored.

Hypothesis	Status after this run
H2 – APC phantom clusters via hash-fallback	STRONGLY CONFIRMED (pre-flight banner + D1 exit 1)
H3 – topk=5 too small	Inconclusive – need the D2 sweep output for $K \in \{5, 10, 15, 20\}$
H10 – sampled-eval vs. all-ranking mismatch	Inconclusive – need the D3 pure-MMHCL output (compare against the 0.0881 paper baseline)

Table 2: Updated status of the three highest-priority hypotheses.

5 Why Did R@20 Still Reach ~0.085 Under Hash-Fallback?

This subtle point deserves a clear explanation. The hash-fallback in `load_data.py` generates clusters as `cluster(i) = hash(item_id) mod num_categories`. The resulting signal has structure but *does not* correlate with the true semantics of any item. When APC consumes it:

- the APC loss still converges – the hash is deterministic, so the clusters remain stable across epochs;
- but the gradient signal propagated into item embeddings is **structured noise** – the embeddings end up learning a “fake feature” whose effect on Recall@20 may help or hurt by chance.

The fact that **combined** (0.0851) is higher than **anchor** (0.0795) demonstrates that static- τ + AVRF-off is *genuinely* useful, because the hash-noise affects both variants symmetrically. But it **cannot** be claimed that 0.0851 is the ceiling of the recipe: restoring the real metadata could push it above 0.087 (passing the stop-gate) or pull it down (if the paper’s APC was tuned for the spurious metadata in the first place).

This also suggests a useful *secondary hypothesis*: if the real metadata raises R@20 substantially, then APC is a genuinely valuable module; if the real metadata barely changes anything, APC is mostly a noise-regulariser and the module can be simplified or removed.

6 Action Items in Priority Order

1. IMMEDIATE – Recover the D2/D3/APC outputs already computed (do not waste 12 h of GPU)

Open a Windows terminal in `MMHCL_DAMPS_Project/` and run:

```
dir /s /b ..\Clothing\*diag_D2_topk*\*.txt
dir /s /b ..\Clothing\*diag_D3_mmhcl_pure*\*.txt
dir /s /b ..\Clothing\*diag_D1_apc_off*\*.txt
```

This should produce 6 `.txt` files (4 D2 + 1 D3 + 1 APC-off). Share their contents, especially the lines:

```
BEST_Test_Recall@20: ...
BEST_Test_NDCG@20: ...
```

Alternatively, scroll back through the cell 15 output and copy **everything** (not just the head and tail). Or, programmatically:

```
import json, nbformat
nb = nbformat.read(
    "Local_Random_seeds_train_mmhcl_clothing_colab_original.ipynb",
    as_version=4,
)
print(nb.cells[15].outputs[0]["text"])
```

2. RESTORE THE METADATA – precondition for a valid Phase 1

Two paths:

- (A) **Download the original file from the MMHCL paper release.** The original MMHCL repository, distributed via the authors’ Google Drive, typically ships `meta_categories.npy` for Clothing, Sports and Baby. Place the file at:

```
data\Clothing\meta_categories.npy
```

Requirements: `len == 23033`, `dtype` is integer, and at least 5 unique cluster labels.

- (B) **Regenerate it from raw Amazon Clothing metadata.** If (A) is unavailable, the file can be regenerated by mapping each `item_id` (range `0...23032`) to its top-level Amazon category (Clothing, Shoes & Jewelry > <subcategory>). A `build_meta_categories.py` script can be provided on request.

3. AFTER METADATA IS RESTORED – re-run the Phase 1 sweep on real clusters

Re-execute the 20 runs of cell 16 (5 seeds $\times$ 4 variants $\times$ 250 epochs, $\approx$ 12 h). Then compare:

- If `combined` reaches $R@20 \geq 0.0870 \Rightarrow$ **stop-gate PASS**; Phase 1 closes and we move on to Phase 2.
- If it still FAILs but $R@20$ changes *substantially* ($\Delta R@20 > 0.003$) $\Rightarrow$ APC carries a real signal; retune `damps_warmup_epochs` and `damps_num_categories`.
- If $R@20$ barely moves ($\Delta R@20 < 0.001$) $\Rightarrow$ APC is essentially noise regularisation; recommend dropping APC and moving directly to tuning K or `rebuild_R`.

4. OPTIONAL – Update markdown cell 14 to record the finding

Add the following note to cell 14:

*The Phase 1 multi-seed sweep (cell 16) was executed with `meta_categories.npy` **MISSING** – all 20 runs used the hash-fallback APC clusters. The stop-gate verdict (cell 20: all variants FAIL) is **confounded by H2** and must be re-evaluated after the metadata file is restored.*

7 Conclusion

Aspect	Assessment
D1 / D2 / D3 infrastructure	Functioning to 100% of spec
D1 verdict	H2 positive – hash-fallback is active
Phase 1 sweep (cell 16)	Contaminated by H2 – stop-gate verdict not final
D2 / D3 / APC outputs	Computed ($\sim$ 12 h) but not yet shared – must be recovered
Critical next step	Restore <code>meta_categories.npy</code> ; re-run the 20-run sweep

Table 3: One-line summary of the current state of Phase 1.

Once the complete cell 15 output (or the contents of the 6 log files under `..\Clothing\diag_*`) is shared, definitive verdicts can be issued for hypotheses **H3** and **H10**, and a precise execution plan for Phase 1 round 2 can be drawn up.